



SUSTAINABLE LIVING APP

TEAM MEMBERS

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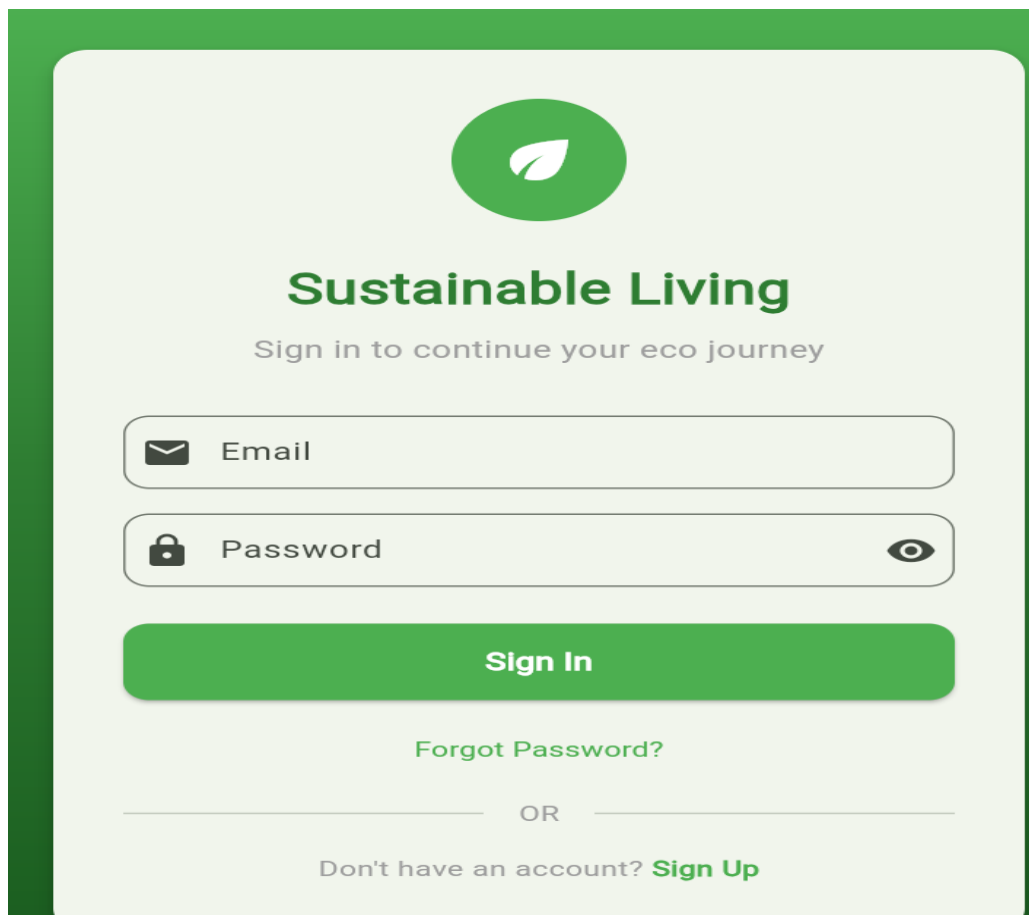
We have successfully built an application on **sustainable living app** as a project of 4th semester [HDSE II].

ACKNOWLEDGEMENT

This project is done by *Asim Ali, Talha Ubaid, M. Bazaan, and M. Usama* of HDSE II [4th Semester] Aptech University Road Centre, Karachi.

The project that we have created has 11 pages named Login, Signup, Home, Challenges, Tips, Carbon Tracker, Profile, Admin Dashboard, Challenge Management, Eco Tips Management, Analytics & Reports.

Login Page UI



The image shows a login page for a web application titled "Sustainable Living". The page has a light green background with a dark green border. At the top center is a circular logo containing a white leaf. Below the logo, the text "Sustainable Living" is displayed in a bold, dark green font. Underneath this, a subtitle reads "Sign in to continue your eco journey". The login form consists of two input fields: "Email" with an envelope icon and "Password" with a lock icon and a toggle eye icon. A prominent green "Sign In" button is located below the password field. Below the button is a link for "Forgot Password?". At the bottom, there is a horizontal line with the word "OR" in the center, and a text prompt "Don't have an account? Sign Up" where "Sign Up" is a green link.



Sustainable Living

Sign in to continue your eco journey

 Email

 Password 

Sign In

[Forgot Password?](#)

OR

Don't have an account? [Sign Up](#)

CODE

```
class LoginScreen extends StatefulWidget {  
  const LoginScreen({super.key});  
  
  @override  
  State<LoginScreen> createState() => _LoginScreenState();  
}  
  
class _LoginScreenState extends State<LoginScreen> {  
  final _formKey = GlobalKey<FormState>();  
  final _emailController = TextEditingController();  
  final _passwordController = TextEditingController();  
  final _authService = AuthService();  
  
  bool _isLoading = false;  
  bool _obscurePassword = true;  
  
  @override  
  void dispose() {  
    _emailController.dispose();  
  }  
}
```

Signup Page UI



Join Sustainable Living

Start your eco-friendly journey today



Full Name



Email



Password



● At least 8 character

CODE

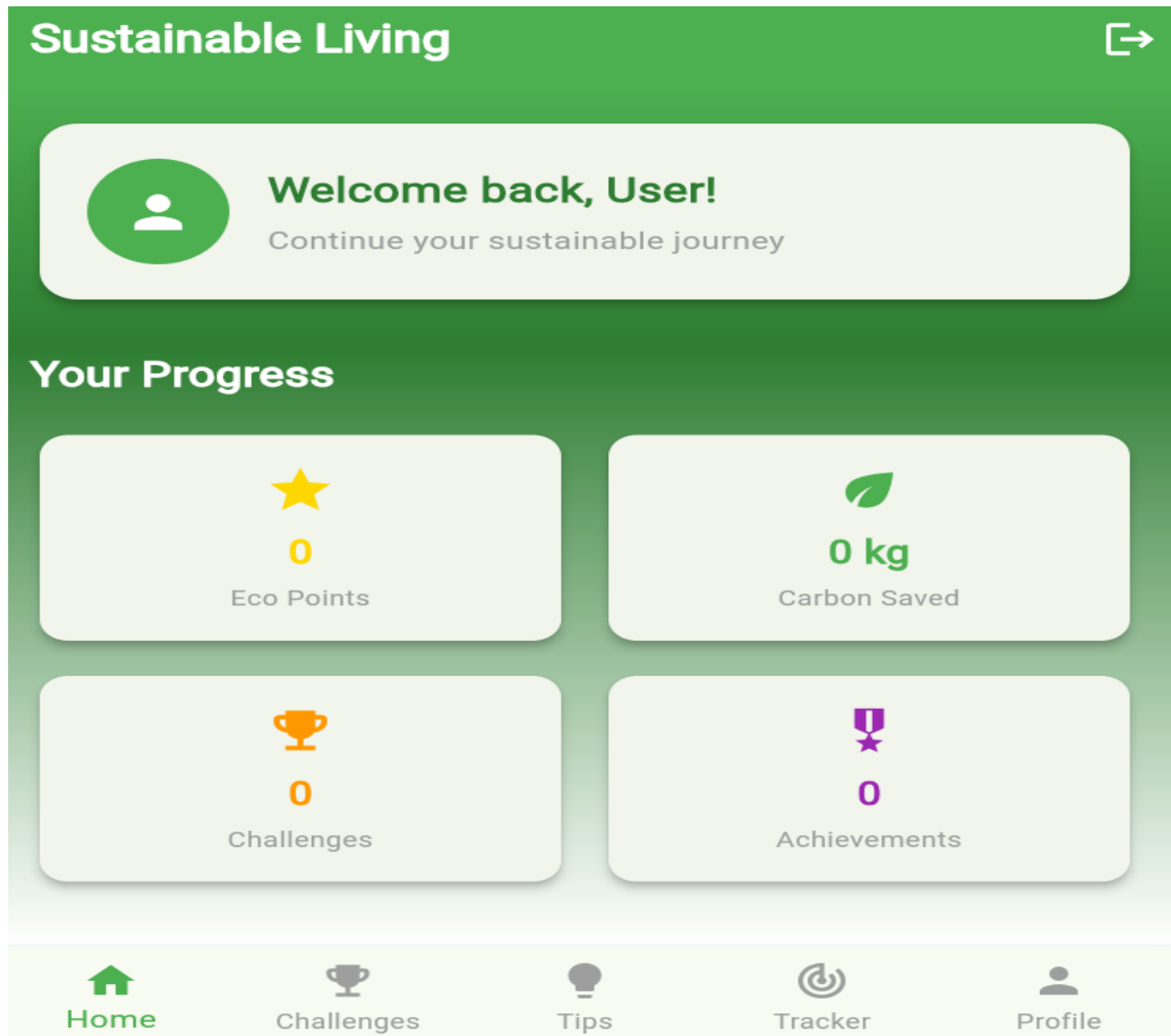
You, 3 weeks ago | 1 author (You) | Windsurf: Refactor | Explain

```
class RegisterScreen extends StatefulWidget {  
  const RegisterScreen({super.key});  
  
  @override  
  State<RegisterScreen> createState() => _RegisterScreenState();  
}
```

You, 3 weeks ago | 1 author (You) | Windsurf: Refactor | Explain

```
class _RegisterScreenState extends State<RegisterScreen> {  
  final _formKey = GlobalKey<FormState>();  
  final _nameController = TextEditingController();  
  final _emailController = TextEditingController();  
  final _passwordController = TextEditingController();  
  final _confirmPasswordController = TextEditingController();  
  final _authService = AuthService();  
  
  bool _isLoading = false;  
  bool _obscurePassword = true;  
  bool _obscureConfirmPassword = true;  
  bool isValidPassword = false;
```

Home Page UI

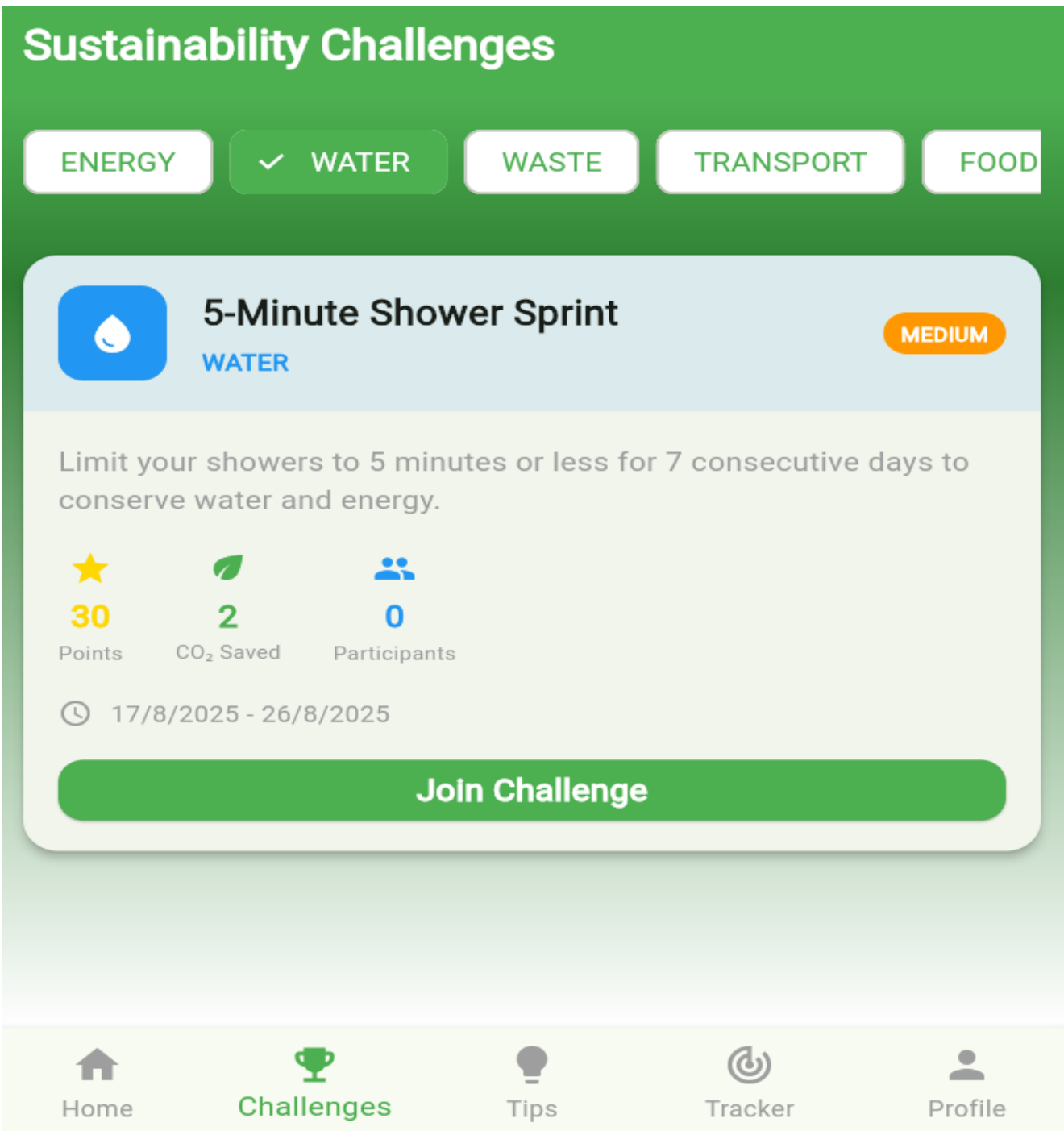


CODE

```
Widget build(BuildContext context) {  
  return Scaffold(  
    appBar: AppBar(  
      title: const Text('Sustainable Living'),  
      backgroundColor:  Colors.deepOrangeAccent,  
    ), // AppBar  
    body: Container(  
      color:  Colors.grey[100],  
      child: SingleChildScrollView(  
        child: Padding(  
          padding: const EdgeInsets.all(20.0),  
          child: Column(  
            crossAxisAlignment: CrossAxisAlignment.start,  
            children: [  
              const Text(  
                'Welcome to Sustainable Living!',  
                style: TextStyle(  
                  fontSize: 26,  
                  fontWeight: FontWeight.bold,  
                  color:  Colors.deepOrangeAccent,  
                ),  
              ),  
            ],  
          ),  
        ),  
      ),  
    ),  
  );  
}
```

You, 3 weeks ago

Challenges Page UI



CODE

@override

Windsurf: Refactor | Explain | Generate Function Comment | ✕

```
Widget build(BuildContext context) {  
  return Scaffold(  
    appBar: AppBar(  
      title: const Text(  
        'Sustainability Challenges',  
        style: TextStyle(  
          color:  Colors.white,  
          fontWeight: FontWeight.bold,  
        ), // TextStyle  
      ), // Text  
      backgroundColor:  const Color(0xFF4CAF50),  
      elevation: 0,  
    ), // AppBar  
    body: Container(  
      decoration: const BoxDecoration(  
        gradient: LinearGradient(  
          begin: Alignment.topCenter,  
          end: Alignment.bottomCenter,  
          colors: [  
            Color(0xFF4CAF50),  
            Color(0xFF4CAF50),  
          ],  
        ),  
      ),  
    ),  
  ),  
);
```

Eco Tips Page UI

Eco Tips & Advice

🔍 Search eco tips...



Meatless Mondays

FOOD

Skip meat once a week to reduce your environmental impact.

Animal agriculture is a major contributor to greenhouse gas emissions, deforestation, and water usage. By choosing plant-based meals just one day a week, you can significantly lower your carbon footprint. It's also a great way to explore new recipes and improve your health.



25

Points



0

Views



0

Likes

🤍 Like

🔗 Share

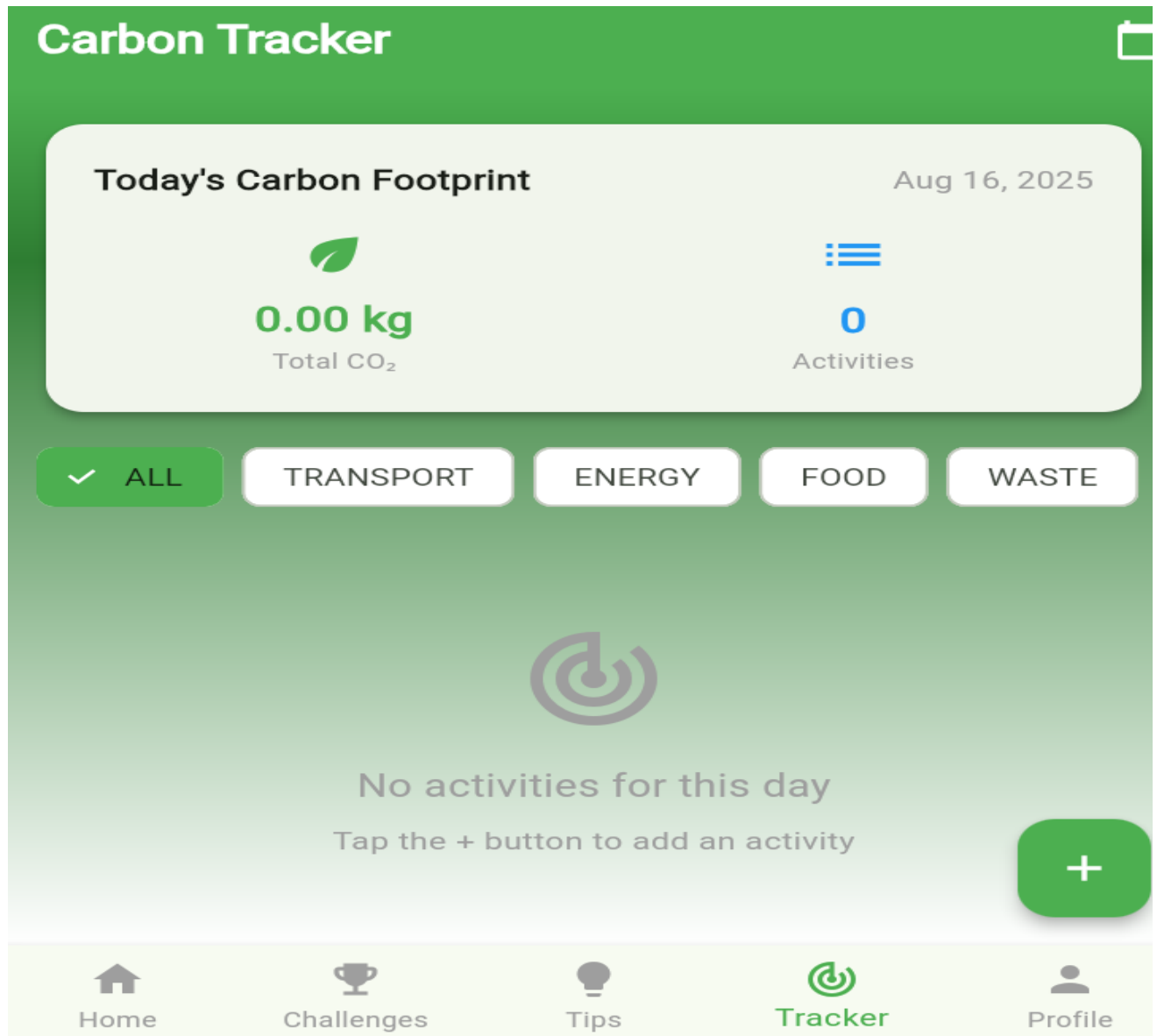
CODE

@override

Windsurf: Refactor | Explain | Generate Function Comment | ✕

```
Widget build(BuildContext context) {  
  return Scaffold(  
    appBar: AppBar(  
      title: const Text(  
        'Eco Tips & Advice',  
        style: TextStyle(  
          color: Colors.white,  
          fontWeight: FontWeight.bold,  
        ), // TextStyle  
      ), // Text  
      backgroundColor: const Color(0xFF4CAF50),  
      elevation: 0,  
    ), // AppBar  
    body: Container(  
      decoration: const BoxDecoration(  
        gradient: LinearGradient(  
          begin: Alignment.topCenter,  
          end: Alignment.bottomCenter,  
          colors: [
```

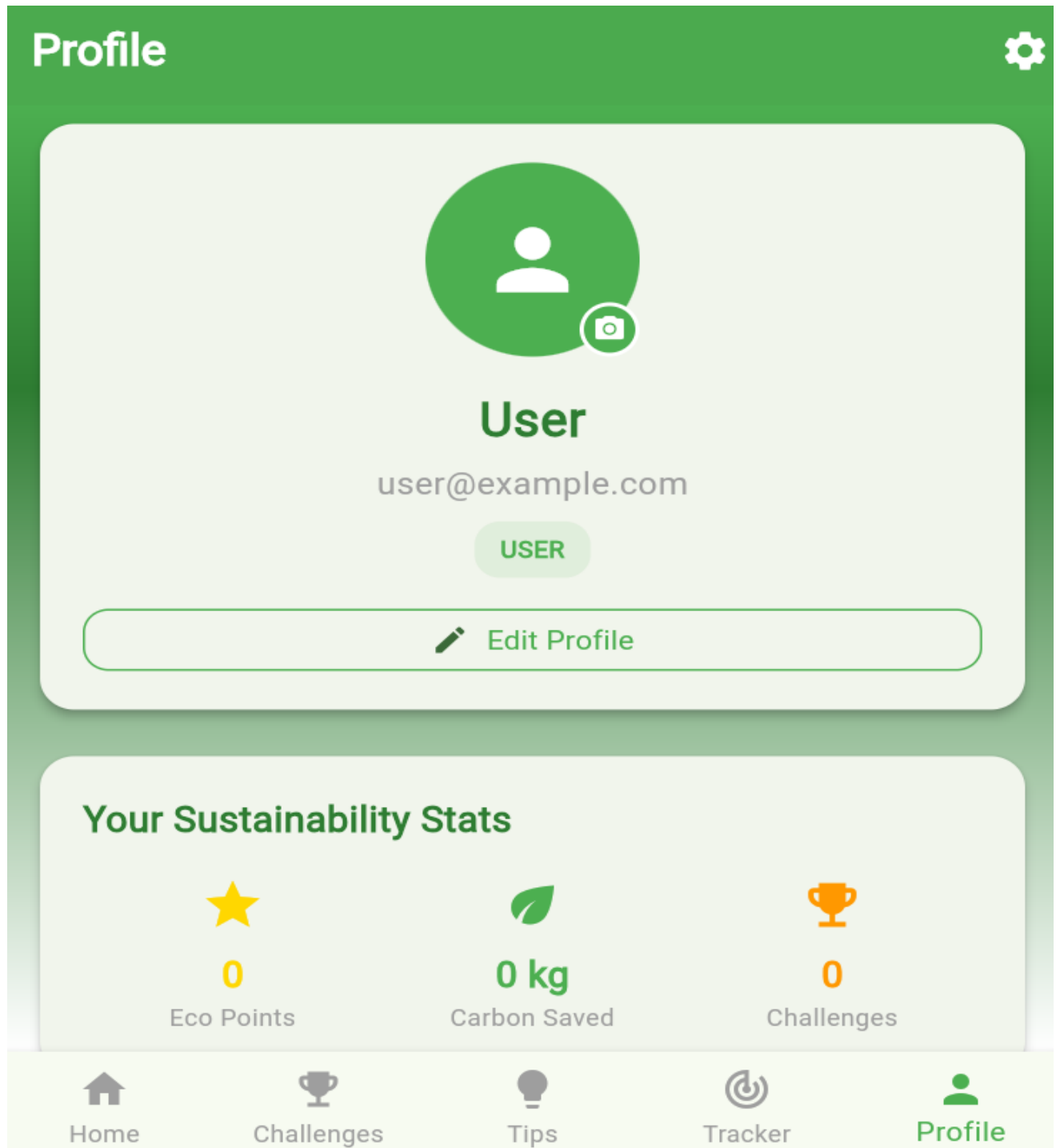
Carbon Tracker Page UI



CODE

```
        color: Colors.white,
        fontWeight: FontWeight.bold,
      ), // TextStyle
    ), // Text
    backgroundColor: const Color(0xFF4CAF50),
    elevation: 0,
    actions: [
      IconButton(
        icon: const Icon(Icons.calendar_today, color: Colors.white),
        onPressed: _selectDate,
      ), // IconButton
    ],
  ), // AppBar
  body: Container(
    decoration: const BoxDecoration(
      gradient: LinearGradient(
        begin: Alignment.topCenter,
        end: Alignment.bottomCenter,
        colors: [
          Color(0xFF4CAF50),
          Color(0xFF2E7D32)
```

Profile Page UI



CODE

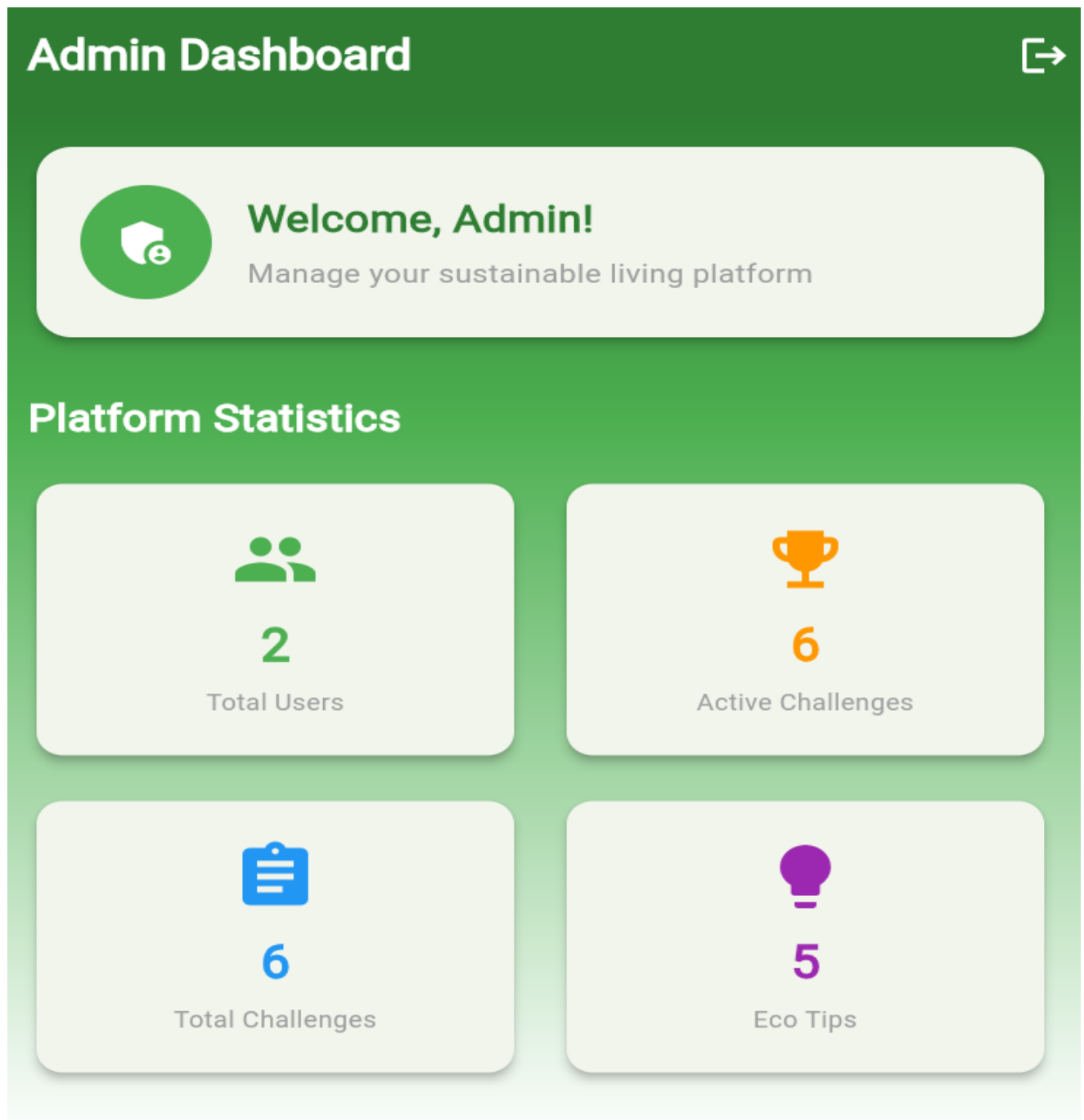
@override

Windsurf: Refactor | Explain | Generate Function Comment | X

```
Widget build(BuildContext context) {  
  return Scaffold(  
    appBar: AppBar(  
      title: const Text(  
        'Profile',  
        style: TextStyle(  
          color: Colors.white,  
          fontWeight: FontWeight.bold,  
        ), // TextStyle  
      ), // Text  
      backgroundColor: const Color(0xFF4CAF50),  
      elevation: 0,  
      actions: [  
        IconButton(  
          icon: const Icon(Icons.settings, color: Colors.white),  
          onPressed: () {  
            Navigator.push(  
              context,  
              MaterialPageRoute(  
                builder: (context) => ProfilePage(),  
              ),  
            );  
          },  
        ),  
      ],  
    ),  
  );  
}
```

Admin Pages

Admin Dashboard UI



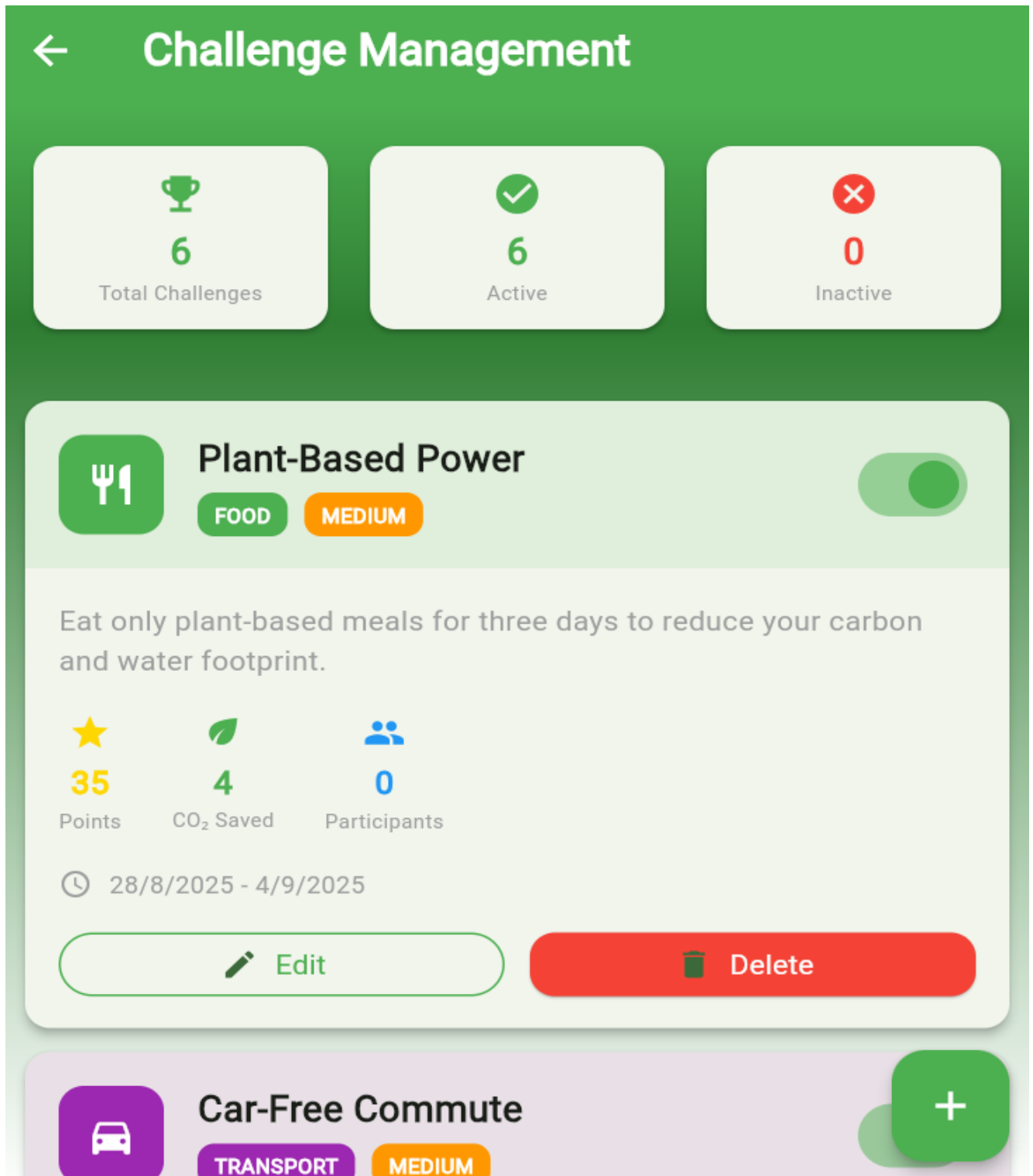
CODE

@override

Windsurf: Refactor | Explain | Generate Function Comment | X

```
Widget build(BuildContext context) {  
  return Scaffold(  
    appBar: AppBar(  
      title: const Text(  
        'Admin Dashboard',  
        style: TextStyle(  
          color: Colors.white,  
          fontWeight: FontWeight.bold,  
        ), // TextStyle  
      ), // Text  
      backgroundColor: const Color(0xFF2E7D32),  
      elevation: 0,  
      actions: [  
        IconButton(  
          icon: const Icon(Icons.logout, color: Colors.white),  
          onPressed: _logout,  
        ), // IconButton  
      ],  
    ),  
  );  
}
```

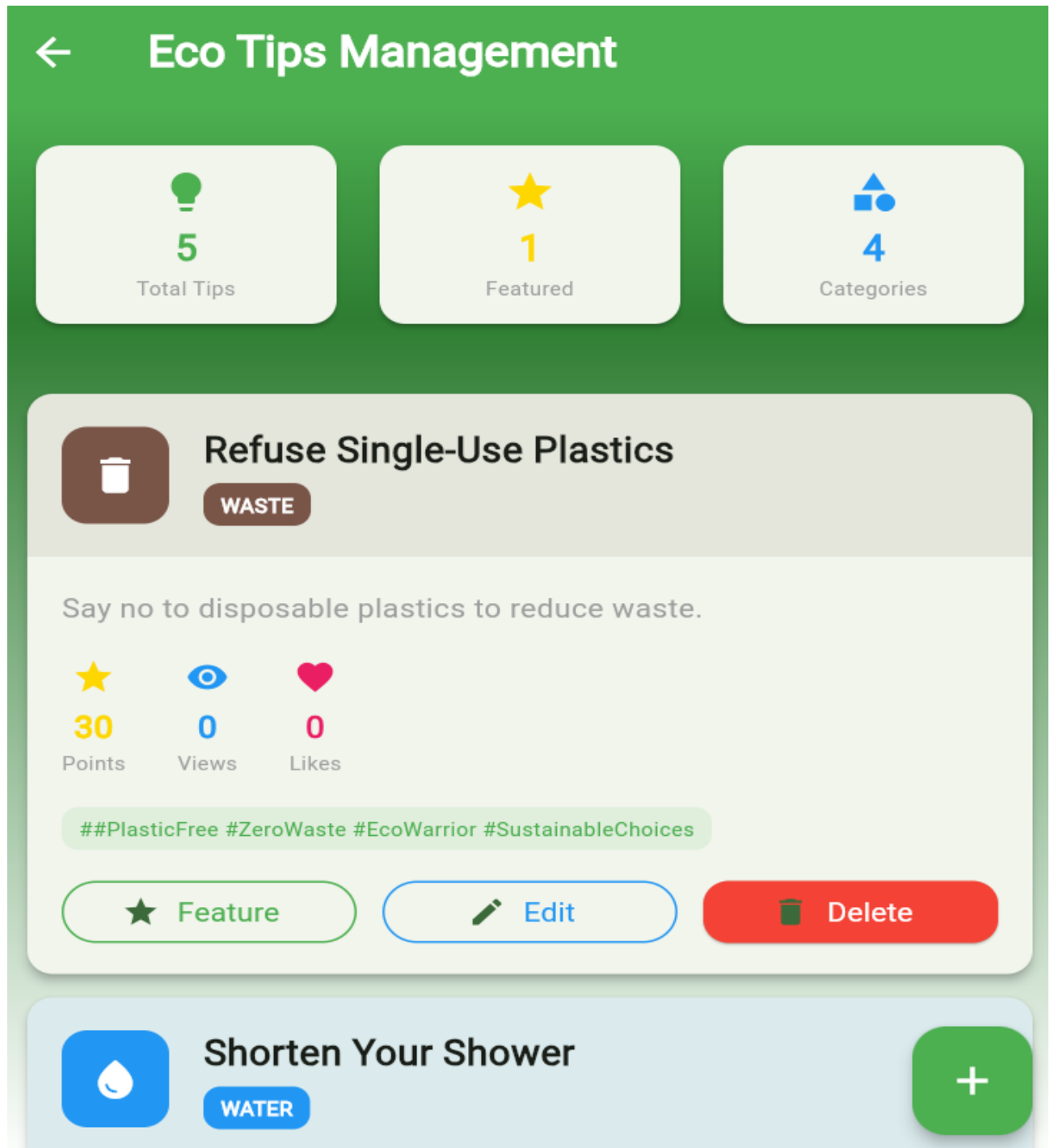
Challenge Management UI



CODE

```
ScaffoldMessenger.of(context).showSnackBar(  
  SnackBar(  
    content: Text(  
      challenge.isActive ? 'Challenge deactivated' : 'Challenge  
        activated',  
    ), // Text  
    backgroundColor:  const Color(0xFF4CAF50),  
  ), // SnackBar  
);  
} catch (e) {  
  ScaffoldMessenger.of(context).showSnackBar(  
    SnackBar(  
      content: Text('Error updating challenge: $e'),  
      backgroundColor:  Colors.red,  
    ), // SnackBar  
  );  
}  
}  
  
Future<void> _deleteChallenge(ChallengeModel challenge) async {  
  // ...  
}
```

Eco Tips Management UI



CODE

@override

Windsurf: Refactor | Explain | Generate Function Comment | ✕

```
Widget build(BuildContext context) {  
  return Scaffold(  
    appBar: AppBar(  
      title: const Text(  
        'Eco Tips Management',  
        style: TextStyle(  
          color: Colors.white,  
          fontWeight: FontWeight.bold,  
        ), // TextStyle  
      ), // Text  
      backgroundColor: const Color(0xFF4CAF50),  
      elevation: 0,  
      leading: IconButton(  
        icon: const Icon(Icons.arrow_back, color: Colors.white),  
        onPressed: () => Navigator.pop(context),  
      ), // IconButton  
    ), // AppBar  
    body: Container(  
      decoration: const BoxDecoration(  
        gradient: LinearGradient(  
          colors: [Colors.white, Colors.white],  
          begin: Alignment.topLeft,  
          end: Alignment.bottomRight,  
          stops: [0.5, 0.5],  
        ),  
      ),  
    ),  
  );  
}
```

Analytics & Reports UI



Analytics & Reports

Platform Overview



0

Total Users



0

Active Challenges



0

Total Eco Points



0

CO₂ Saved (kg)

User Engagement



CODE

@override

Windsurf: Refactor | Explain | Generate Function Comment | X

```
Widget build(BuildContext context) {  
  return Scaffold(  
    appBar: AppBar(  
      title: const Text(  
        'Analytics & Reports',  
        style: TextStyle(  
          color: Colors.white,  
          fontWeight: FontWeight.bold,  
        ), // TextStyle  
      ), // Text  
      backgroundColor: const Color(0xFF4CAF50),  
      elevation: 0,  
      leading: IconButton(  
        icon: const Icon(Icons.arrow_back, color: Colors.white),  
        onPressed: () => Navigator.pop(context),  
      ), // IconButton  
    ), // AppBar  
    body: _isLoading  
      ? const Center(  
        child: CircularProgressIndicator(),  
      )  
      : const Center(  
        child: Text('Loading...'),  
      )  
  );  
}
```

PROBLEM STATEMENT

1. Introduction

1.1 Background and Necessity for the Mobile App

As awareness of climate change, environmental degradation, and resource depletion increases, many people are looking for ways to live more sustainably. However, the complexity of eco-friendly living—ranging from reducing waste and conserving energy to making ethical purchasing decisions—can overwhelm individuals who want to make a difference. While there is a wealth of information available on sustainable living, people often struggle to know where to begin or how to integrate these practices into their everyday routines.

In response to this need, there is a growing demand for tools that simplify sustainable living, helping individuals track their environmental impact and make informed, responsible choices. An app that educates users on sustainable practices, suggests eco-friendly products, and tracks their carbon footprint can empower users to make meaningful changes, while offering a personalized roadmap for adopting a greener lifestyle.

By providing the tools, information, and resources to make sustainable living easier and more accessible, the Sustainable Living Guide app will empower users to reduce their environmental impact and make more eco-conscious choices. Through tracking, education, and personalized recommendations, the app will help individuals live in a way that aligns with their values and contributes positively to the planet's future.

1.2 Proposed Solution

The Sustainable Living Guide app will serve as an all-in-one platform to educate users on sustainable living, provide eco-friendly product recommendations, and offer carbon footprint tracking tools. The app will guide users through various aspects of their daily lives, from sustainable shopping to waste reduction, energy conservation, and conscious transportation choices. By tracking and providing actionable insights, the app will help users measure their impact and make positive changes that contribute to a more sustainable future.

1.3 Purpose of the Document

This document outlines the functional and non-functional requirements, technical specifications, and constraints of the Sustainable Living Guide App. It serves as a guideline for developers, designers, and stakeholders involved in the project.

1.4 Constraints

The development of the Sustainable Living Guide App must adhere to several constraints to ensure its successful implementation and operation. You may encounter constraints related to data storage, data synchronization, and backup procedures. This mobile app will be displaying various images and information, hence loading time should be minimized and fetching of data within the app should be fast and seamless. The usage of various images and videos may be subject to licensing agreements and copyright restrictions. It is important to understand and comply with these constraints to avoid legal issues

2. Functional Requirements

The app will be designed as a set of forms/pages, Navigation, and Fragments/Screens with menus representing choice of activities to be performed.

Following are the functional requirements of the mobile app:

1. **Carbon Footprint Tracker:** Users can input daily activities (e.g., transportation, energy usage, food consumption) to calculate their carbon footprint. The app will provide insights into the user's carbon impact and offer recommendations to reduce it.
2. **Eco-Friendly Product Suggestions:** Based on the user's preferences and lifestyle, the app will suggest sustainable, eco-friendly products. This could include items like reusable water bottles, organic food options, energy-efficient appliances, and zero-waste alternatives.
3. **Sustainable Living Challenges:** The app will feature weekly or monthly challenges to encourage users to adopt specific sustainable habits (e.g., a "Plastic-Free Week").
4. **Green Certifications & Labels Guide:** The app will educate users on certifications and eco-labels (e.g., Fair Trade, Energy Star, Organic) to help them identify genuinely sustainable products and services when shopping.
5. **Waste Reduction Tracker:** A feature to help users track their waste reduction efforts, including recycling habits, composting, and reducing single-use plastics. The app will provide tips and statistics to motivate users to improve their waste management.

6. **Sustainable Recipes & Meal Planning:** The app will offer eco-friendly meal planning suggestions that focus on plant-based, locally sourced, and organic ingredients. This can help users reduce their food-related carbon footprint by making better food choices.
7. **Energy Conservation Tips:** The app will provide users with personalized energy-saving tips for their home, such as turning off unused appliances, optimizing heating and cooling, or using renewable energy sources.
8. **Eco-Travel Suggestions:** Recommendations for eco-friendly travel options, such as public transportation, carpooling, and sustainable hotels. The app will also offer sustainable travel tips to reduce the environmental impact of vacations and trips.
9. **Community Forum & Support:** A platform where users can share their sustainability efforts, challenges, and success stories. This community aspect will foster a sense of support and inspiration for users as they embark on their eco-friendly journeys.
10. **Sustainability Progress Dashboard:** A personalized dashboard that tracks a user's sustainable living progress over time. Users can see improvements in areas like waste reduction, carbon footprint, and sustainable purchasing habits.
11. **Educational Content:** Articles, videos, and infographics that provide users with information on environmental issues and sustainable practices. This content will help users stay informed and inspired to make positive changes.